# Mikhail Razakov

## Compiler / Software Protection Engineer

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I work on symbolic computation in SharpChecker at ISP RAS. Public projects cover LLVM analysis and transformations, x86-64 codegen, and Roslyn CFG. Software-protection evidence is limited to a historical Unity licensing prototype with server-side key activation and activation limits.

## PROFESSIONAL EXPERIENCE

**October 2026 — present**

### ISP RAS — SharpChecker development

- Work on symbolic computation in SharpChecker, a C#/.NET static-analysis tool.

**July — August 2026**

### MCST — Compiler Engineering Intern

- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
- Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.

**Independent work (not employment)** — Unity licensing prototype: server-side key activation and limits; VpnMediator: subscription/access state and recovery. Neither project is represented as an employed role.

## PROJECTS

### LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv,verify, structural expectations, idempotence, and observable before/after execution comparison.

### x86-64 Compiler Backend & Register Allocation

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

### DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.

## SKILLS

**Compiler / LLVM**

C++23, LLVM 22/IR, interprocedural analysis, legality, transformations, LLVM Verifier

**Program Analysis**

Roslyn CFG, fixed-point data flow, state propagation/joins, conservative analysis

**Low-Level / Codegen**

Rust, SysV x86-64, liveness/interference, register allocation, spills, phi lowering

**Software Protection / Licensing**

server-side activation, activation-count limits, Unity integration

## EDUCATION

HSE University — Software Engineering

## RECOGNITION

**MCST LLVM/C++ selection**

Organizer correspondence: only 5.0/5.0 score and only solution passing all 104/104 tests among 49 submissions.

**LangDev'26 — Build the Language, Then Make the Abstractions Disappear**

Delivered the UniversalToolchain architecture talk at LangDev on 9 October 2026.

**HSE Faculty of Computer Science — open-source projects competition**

Selected as one of the winners with UniversalToolchain / Wist2.

**Earlier distinctions**

MEPhI Junior: Information Technologies section winner in 2025 and first-degree diploma in 2026; HSE Vysshaya Proba Industrial Programming diploma holder; Baltic Science and Engineering Competition: first-degree diploma and the "Perfection as Hope" prize.

Moscow / remote